B. SYLLABUS

COURSE NO	TITLE OF THE COURSE	COURSE STRUCTURE	PRE-REQUISITE
FCCSE002	Computer Programming	3L - 0T - 2P	None

COURSE OUTCOMES (COs)

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills, and attitudes:

1. To identify and understand the syntax and features of Python.
2. To utilize control structures and practice the Python concepts like classes, objects & functions to solve programming problems.
3. To use arrays, strings and other data structures and to compare and contrast among different python libraries.
4. To implement file handling operations and perform input/output operations.
5. To design object-oriented Python programs and apply Python programming concepts to real-world scenarios.

COURSE CONTENTS

Unit I

Introduction to Python Programming: Introduction to Python and Environment Setup, Python Syntax and Basic Data Types, Variables, Operators, and Expressions, Control Structures and Conditional Statements, Loops and Iterations: For, While, For-else, Range function Sentinel-Controlled Iteration , break and continue Statements, Defining Functions, Functions with Multiple Parameters, Default Parameter Values, Keyword Arguments, Arbitrary Argument Lists, Lists and Tuples, Associated in-built functions, Dictionaries and Sets

Unit II

Object-Oriented Programming in Python: Object-Oriented Programming (OOP) Fundamentals, Classes and Objects, Inheritance and Polymorphism, Advanced OOP Concepts (e.g., Encapsulation, Abstraction), Generators and Iterators, Functional Programming Techniques (e.g., Lambda Functions, Map, Filter), Modules and Packages

Unit III

Arrays and Strings: Introduction to NumPy Arrays, Array Operations and Functions, Indexing and slicing arrays, Array Manipulation and Stacking, multidimensional arrays and matrices, Creating and manipulating strings, String Operations, Regular Expressions, pattern matching and substitution, Working with Dates and Times, Introduction to pandas Series and DataFrames, Data Analysis with Pandas (e.g., Filtering, Grouping, Aggregation)

Unit IV

File and Exception handling: File handling in Python, Different types of files and their uses, Updating text files: appending and modifying data, Understanding serialization and deserialization, Working with JSON files for data storage and retrieval, Handling Exceptions using try and except statements, Catching and handling specific exceptions, Raising custom exceptions, 'finally' clause for cleanup operations, Explicitly raising exceptions with 'raise', Working with CSV files

Unit V

Web Development and data analysis using python: Introduction to Web Development with Python, Introduction to Flask Framework, defining routes and handling HTTP methods (GET, POST), Request and response objects in Flask, Building Web Applications with Flask, Introduction to Data Visualization, Data Visualization with Matplotlib

SUGGESTED READINGS

1. R. Nageswara Rao, "**Core Python Programming**" Dreamtech Press, 3^{rd} Edition, 2021.
2. Martin C. Brown, "**Python: The Complete Reference**" McGraw Hill Publications, 1^{st} Edition, 2021.
3. Paul J. Deitel, Harvey Deitel "**Python for Programmers**", Pearson publications, 1^{st} Edition, 2020.
4. Yashavant Kanetkar, "Let Us Python". BPB Publications, 5^{th} edition,2023.